CAD Standards Library

Industry-Grade Drafting & Modeling Standards for Engineers

Purpose of the CAD Standards Library

Inconsistent CAD practices cost companies thousands of hours in rework, miscommunication, and manufacturing errors. The JC Tech CAD Standards Library provides a unified set of rules, templates, and best practices for 2D drafting and 3D modeling that align with international standards (ISO 128, ASME Y14.5, BIS SP 46) and industry best practices from automotive, aerospace, and heavy machinery sectors.

What's Included

- Drawing Templates: Pre-configured title blocks, border layouts, and layer standards for AutoCAD, SolidWorks, CATIA, and Creo.
- GD&T Quick Reference: Complete symbol library with application rules per ASME Y14.5-2018 and ISO 1101.
- Dimensioning Standards: Rules for linear, angular, radial, and ordinate dimensioning with tolerance callouts.
- Part Numbering Systems: Intelligent and sequential part numbering conventions for assembly management.
- Revision Control: Standards for revision blocks, ECN (Engineering Change Notice) workflows, and version tracking.
- 3D Modeling Best Practices: Feature tree organization, reference geometry naming, multi-body vs. assembly strategies.

Supported CAD Platforms

2D Drafting:	AutoCAD, DraftSight, LibreCAD
3D Parametric:	SolidWorks, CATIA V5/V6, Creo Parametric, Siemens NX

Simulation:	ANSYS Mechanical, Abaqus, SimScale
Data Exchange:	STEP (AP214/AP242), IGES, JT, 3D PDF

Sample Standard: Drawing Layer Convention

0-GEOMETRY:	Primary part geometry (visible edges) -- Color: White, Lineweight: 0.35mm
1-HIDDEN:	Hidden lines -- Color: Grey (8), Linetype: HIDDEN, Lineweight: 0.18mm
2-CENTER:	Center lines and axes -- Color: Red, Linetype: CENTER, Lineweight: 0.18mm
3-DIMENSION:	All dimensions and tolerances -- Color: Green, Lineweight: 0.25mm
4-HATCH:	Section hatching and fill patterns -- Color: Yellow, Lineweight: 0.18mm
5-TEXT:	Notes, callouts, and annotations -- Color: Cyan, Lineweight: 0.25mm

Why Standards Matter in CAD

A 2023 industry survey revealed that 67% of manufacturing defects traceable to design originated from ambiguous or non-standard drawings. CAD standards eliminate interpretation errors, speed up design reviews, enable seamless collaboration between teams and suppliers, and ensure regulatory compliance.

Implementation Guide

- Step 1: Download the template pack for your CAD platform from the JC Tech Knowledge Hub.
- Step 2: Import templates into your CAD environment and set as default for new projects.
- Step 3: Train your team using the included quick-reference cards and video walkthroughs.
- Step 4: Establish a review checklist to audit drawings against the standard before release.

Download & Support

All templates and reference documents are available for free download. Enterprise clients can request customized standards aligned with their internal processes.

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